# ŘEŠENÍ Bonus 2

① a)
$$\left(\begin{array}{ccc|c} \textcircled{1} & -2 & 1 & 1 \\ 2 & 1 & -1 & 2 \\ 1 & 1 & -2 & 3 \end{array}\right) \begin{matrix} \\ \cdot(-2) \\ \cdot(-1) \end{matrix} \sim \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & \textcircled{5} & -3 & 0 \\ 0 & 3 & -3 & 2 \end{array}\right) \begin{matrix} \\ \cdot(-3) \\ \cdot 5 \end{matrix}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -2 & 1 & 1 \\ 0 & 5 & -3 & 0 \\ 0 & 0 & -6 & 10 \end{array}\right)$$

$-6x_3 = 10$ $\quad$ $5x_2 - 3x_3 = 0$ $\quad$ $x_1 - 2x_2 + x_3 = 1$

$x_3 = -\frac{5}{3}$ $\quad$ $5x_2 + 5 = 0$ $\quad$ $x_1 + 2 - \frac{5}{3} = 1$

$x_2 = -1$ $\quad$ $x_1 = \frac{2}{3}$

b)
$$\begin{pmatrix} 1 & -2 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

② $$\begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} = (2-\lambda)(4-\lambda) - 3 = \lambda^2 - 6\lambda + 5 = (\lambda - 1)(\lambda - 5)$$

$\lambda_1 = 1$ , $\lambda_2 = 5$

VLASTNÍ VEKTORY:

$\lambda_1 = 1$ : $\begin{pmatrix} 1 & 3 \\ 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow u_1 + 3u_2 = 0 \Rightarrow u_1 = -3u_2 \Rightarrow \begin{pmatrix} -3 \\ 1 \end{pmatrix}$

$\lambda_2 = 5$ : $\begin{pmatrix} -3 & 3 \\ 1 & -1 \end{pmatrix} \cdot \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow u_1 - u_2 = 0 \Rightarrow u_1 = u_2 \Rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}$